

Table 6: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.05$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0604 (0.0034) | 0.0496 (0.0031) | 0.0414 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0546 (0.0032) | 0.0544 (0.0032) | 0.0452 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0556 (0.0032) | 0.0592 (0.0033) | 0.0434 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0538 (0.0032) | 0.0554 (0.0032) | 0.0410 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0418 (0.0028) | 0.0544 (0.0032) | 0.0422 (0.0028) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0604 (0.0034) | 0.0496 (0.0031) | 0.0414 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0546 (0.0032) | 0.0544 (0.0032) | 0.0452 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0556 (0.0032) | 0.0592 (0.0033) | 0.0434 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0538 (0.0032) | 0.0554 (0.0032) | 0.0410 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0418 (0.0028) | 0.0544 (0.0032) | 0.0422 (0.0028) |

Table 7: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.05$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0592 (0.0033) | 0.0550 (0.0032) | 0.0346 (0.0026) |
| 100     | 0.0000 (0.0000) | 0.0626 (0.0034) | 0.0524 (0.0032) | 0.0346 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0496 (0.0031) | 0.0586 (0.0033) | 0.0372 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0506 (0.0031) | 0.0536 (0.0032) | 0.0416 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0400 (0.0028) | 0.0536 (0.0032) | 0.0438 (0.0029) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0592 (0.0033) | 0.0550 (0.0032) | 0.0346 (0.0026) |
| 100     | 0.0000 (0.0000) | 0.0626 (0.0034) | 0.0524 (0.0032) | 0.0346 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0496 (0.0031) | 0.0586 (0.0033) | 0.0372 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0506 (0.0031) | 0.0536 (0.0032) | 0.0416 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0400 (0.0028) | 0.0536 (0.0032) | 0.0438 (0.0029) |

Table 8: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.05$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0606 (0.0034) | 0.0524 (0.0032) | 0.0250 (0.0022) |
| 100     | 0.0000 (0.0000) | 0.0576 (0.0033) | 0.0562 (0.0033) | 0.0370 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0506 (0.0031) | 0.0660 (0.0035) | 0.0310 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0546 (0.0032) | 0.0522 (0.0031) | 0.0298 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0390 (0.0027) | 0.0630 (0.0034) | 0.0320 (0.0025) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0606 (0.0034) | 0.0524 (0.0032) | 0.0250 (0.0022) |
| 100     | 0.0000 (0.0000) | 0.0576 (0.0033) | 0.0562 (0.0033) | 0.0370 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0506 (0.0031) | 0.0660 (0.0035) | 0.0310 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0546 (0.0032) | 0.0522 (0.0031) | 0.0298 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0390 (0.0027) | 0.0630 (0.0034) | 0.0320 (0.0025) |

Table 9: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.05$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0616 (0.0034) | 0.0474 (0.0030) | 0.0182 (0.0019) |
| 100     | 0.0000 (0.0000) | 0.0642 (0.0035) | 0.0552 (0.0032) | 0.0224 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0486 (0.0030) | 0.0642 (0.0035) | 0.0316 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0538 (0.0032) | 0.0568 (0.0033) | 0.0324 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0468 (0.0030) | 0.0630 (0.0034) | 0.0302 (0.0024) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0616 (0.0034) | 0.0474 (0.0030) | 0.0182 (0.0019) |
| 100     | 0.0000 (0.0000) | 0.0642 (0.0035) | 0.0552 (0.0032) | 0.0224 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0486 (0.0030) | 0.0642 (0.0035) | 0.0316 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0538 (0.0032) | 0.0568 (0.0033) | 0.0324 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0468 (0.0030) | 0.0630 (0.0034) | 0.0302 (0.0024) |

Table 10: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.1$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0558 (0.0032) | 0.0474 (0.0030) | 0.0402 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0564 (0.0033) | 0.0516 (0.0031) | 0.0428 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0556 (0.0032) | 0.0614 (0.0034) | 0.0344 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0532 (0.0032) | 0.0482 (0.0030) | 0.0390 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0428 (0.0029) | 0.0580 (0.0033) | 0.0474 (0.0030) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0558 (0.0032) | 0.0474 (0.0030) | 0.0402 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0564 (0.0033) | 0.0516 (0.0031) | 0.0428 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0556 (0.0032) | 0.0614 (0.0034) | 0.0344 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0532 (0.0032) | 0.0482 (0.0030) | 0.0390 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0428 (0.0029) | 0.0580 (0.0033) | 0.0474 (0.0030) |

Table 11: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.1$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0608 (0.0034) | 0.0488 (0.0030) | 0.0324 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0598 (0.0034) | 0.0554 (0.0032) | 0.0322 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0502 (0.0031) | 0.0568 (0.0033) | 0.0392 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0522 (0.0031) | 0.0390 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0422 (0.0028) | 0.0654 (0.0035) | 0.0354 (0.0026) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0608 (0.0034) | 0.0488 (0.0030) | 0.0324 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0598 (0.0034) | 0.0554 (0.0032) | 0.0322 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0502 (0.0031) | 0.0568 (0.0033) | 0.0392 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0522 (0.0031) | 0.0390 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0422 (0.0028) | 0.0654 (0.0035) | 0.0354 (0.0026) |

Table 12: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.1$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0594 (0.0033) | 0.0554 (0.0032) | 0.0264 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0508 (0.0031) | 0.0568 (0.0033) | 0.0268 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0508 (0.0031) | 0.0634 (0.0034) | 0.0310 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0504 (0.0031) | 0.0518 (0.0031) | 0.0234 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0442 (0.0029) | 0.0630 (0.0034) | 0.0358 (0.0026) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0594 (0.0033) | 0.0554 (0.0032) | 0.0264 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0508 (0.0031) | 0.0568 (0.0033) | 0.0268 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0508 (0.0031) | 0.0634 (0.0034) | 0.0310 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0504 (0.0031) | 0.0518 (0.0031) | 0.0234 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0442 (0.0029) | 0.0630 (0.0034) | 0.0358 (0.0026) |

Table 13: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.1$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0680 (0.0036) | 0.0470 (0.0030) | 0.0234 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0648 (0.0035) | 0.0612 (0.0034) | 0.0210 (0.0020) |
| 200     | 0.0000 (0.0000) | 0.0498 (0.0031) | 0.0642 (0.0035) | 0.0240 (0.0022) |
| 500     | 0.0000 (0.0000) | 0.0544 (0.0032) | 0.0588 (0.0033) | 0.0258 (0.0022) |
| 1000    | 0.0000 (0.0000) | 0.0428 (0.0029) | 0.0608 (0.0034) | 0.0262 (0.0023) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0680 (0.0036) | 0.0470 (0.0030) | 0.0234 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0648 (0.0035) | 0.0612 (0.0034) | 0.0210 (0.0020) |
| 200     | 0.0000 (0.0000) | 0.0498 (0.0031) | 0.0642 (0.0035) | 0.0240 (0.0022) |
| 500     | 0.0000 (0.0000) | 0.0544 (0.0032) | 0.0588 (0.0033) | 0.0258 (0.0022) |
| 1000    | 0.0000 (0.0000) | 0.0428 (0.0029) | 0.0608 (0.0034) | 0.0262 (0.0023) |

Table 14: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.2$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0582 (0.0033) | 0.0544 (0.0032) | 0.0466 (0.0030) |
| 100     | 0.0000 (0.0000) | 0.0578 (0.0033) | 0.0516 (0.0031) | 0.0376 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0472 (0.0030) | 0.0574 (0.0033) | 0.0454 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0526 (0.0032) | 0.0530 (0.0032) | 0.0360 (0.0026) |
| 1000    | 0.0000 (0.0000) | 0.0444 (0.0029) | 0.0624 (0.0034) | 0.0428 (0.0029) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0582 (0.0033) | 0.0544 (0.0032) | 0.0466 (0.0030) |
| 100     | 0.0000 (0.0000) | 0.0578 (0.0033) | 0.0516 (0.0031) | 0.0376 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0472 (0.0030) | 0.0574 (0.0033) | 0.0454 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0526 (0.0032) | 0.0530 (0.0032) | 0.0360 (0.0026) |
| 1000    | 0.0000 (0.0000) | 0.0444 (0.0029) | 0.0624 (0.0034) | 0.0428 (0.0029) |

Table 15: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.2$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0520 (0.0031) | 0.0488 (0.0030) | 0.0316 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0526 (0.0032) | 0.0520 (0.0031) | 0.0300 (0.0024) |
| 200     | 0.0000 (0.0000) | 0.0470 (0.0030) | 0.0616 (0.0034) | 0.0376 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0498 (0.0031) | 0.0560 (0.0033) | 0.0368 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0368 (0.0027) | 0.0606 (0.0034) | 0.0340 (0.0026) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0520 (0.0031) | 0.0488 (0.0030) | 0.0316 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0526 (0.0032) | 0.0520 (0.0031) | 0.0300 (0.0024) |
| 200     | 0.0000 (0.0000) | 0.0470 (0.0030) | 0.0616 (0.0034) | 0.0376 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0498 (0.0031) | 0.0560 (0.0033) | 0.0368 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0368 (0.0027) | 0.0606 (0.0034) | 0.0340 (0.0026) |

Table 16: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.2$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0516 (0.0031) | 0.0568 (0.0033) | 0.0282 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0496 (0.0031) | 0.0536 (0.0032) | 0.0270 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0492 (0.0031) | 0.0608 (0.0034) | 0.0290 (0.0024) |
| 500     | 0.0000 (0.0000) | 0.0494 (0.0031) | 0.0574 (0.0033) | 0.0322 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0408 (0.0028) | 0.0648 (0.0035) | 0.0300 (0.0024) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0516 (0.0031) | 0.0568 (0.0033) | 0.0282 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0496 (0.0031) | 0.0536 (0.0032) | 0.0270 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0492 (0.0031) | 0.0608 (0.0034) | 0.0290 (0.0024) |
| 500     | 0.0000 (0.0000) | 0.0494 (0.0031) | 0.0574 (0.0033) | 0.0322 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0408 (0.0028) | 0.0648 (0.0035) | 0.0300 (0.0024) |

Table 17: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.25$ ,  $\eta = 0.2$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0518 (0.0031) | 0.0572 (0.0033) | 0.0234 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0516 (0.0031) | 0.0638 (0.0035) | 0.0224 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0462 (0.0030) | 0.0672 (0.0035) | 0.0182 (0.0019) |
| 500     | 0.0000 (0.0000) | 0.0506 (0.0031) | 0.0548 (0.0032) | 0.0226 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0680 (0.0036) | 0.0266 (0.0023) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0518 (0.0031) | 0.0572 (0.0033) | 0.0234 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0516 (0.0031) | 0.0638 (0.0035) | 0.0224 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0462 (0.0030) | 0.0672 (0.0035) | 0.0182 (0.0019) |
| 500     | 0.0000 (0.0000) | 0.0506 (0.0031) | 0.0548 (0.0032) | 0.0226 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0680 (0.0036) | 0.0266 (0.0023) |

Table 18: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.05$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0478 (0.0030) | 0.0540 (0.0032) | 0.0360 (0.0026) |
| 100     | 0.0000 (0.0000) | 0.0472 (0.0030) | 0.0552 (0.0032) | 0.0414 (0.0028) |
| 200     | 0.0000 (0.0000) | 0.0382 (0.0027) | 0.0558 (0.0032) | 0.0428 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0524 (0.0032) | 0.0556 (0.0032) | 0.0374 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0384 (0.0027) | 0.0588 (0.0033) | 0.0400 (0.0028) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0478 (0.0030) | 0.0540 (0.0032) | 0.0360 (0.0026) |
| 100     | 0.0000 (0.0000) | 0.0472 (0.0030) | 0.0552 (0.0032) | 0.0414 (0.0028) |
| 200     | 0.0000 (0.0000) | 0.0382 (0.0027) | 0.0558 (0.0032) | 0.0428 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0524 (0.0032) | 0.0556 (0.0032) | 0.0374 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0384 (0.0027) | 0.0588 (0.0033) | 0.0400 (0.0028) |

Table 19: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.05$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0446 (0.0029) | 0.0478 (0.0030) | 0.0380 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0486 (0.0030) | 0.0548 (0.0032) | 0.0414 (0.0028) |
| 200     | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0602 (0.0034) | 0.0364 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0458 (0.0030) | 0.0524 (0.0032) | 0.0352 (0.0026) |
| 1000    | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0592 (0.0033) | 0.0354 (0.0026) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0446 (0.0029) | 0.0478 (0.0030) | 0.0380 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0486 (0.0030) | 0.0548 (0.0032) | 0.0414 (0.0028) |
| 200     | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0602 (0.0034) | 0.0364 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0458 (0.0030) | 0.0524 (0.0032) | 0.0352 (0.0026) |
| 1000    | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0592 (0.0033) | 0.0354 (0.0026) |

Table 20: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.05$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0506 (0.0031) | 0.0314 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0420 (0.0028) | 0.0536 (0.0032) | 0.0340 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0356 (0.0026) | 0.0700 (0.0036) | 0.0338 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0416 (0.0028) | 0.0530 (0.0032) | 0.0332 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0356 (0.0026) | 0.0688 (0.0036) | 0.0304 (0.0024) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0506 (0.0031) | 0.0314 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0420 (0.0028) | 0.0536 (0.0032) | 0.0340 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0356 (0.0026) | 0.0700 (0.0036) | 0.0338 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0416 (0.0028) | 0.0530 (0.0032) | 0.0332 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0356 (0.0026) | 0.0688 (0.0036) | 0.0304 (0.0024) |

Table 21: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.05$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0396 (0.0028) | 0.0538 (0.0032) | 0.0194 (0.0020) |
| 100     | 0.0000 (0.0000) | 0.0424 (0.0028) | 0.0626 (0.0034) | 0.0264 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0678 (0.0036) | 0.0320 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0474 (0.0030) | 0.0578 (0.0033) | 0.0282 (0.0023) |
| 1000    | 0.0000 (0.0000) | 0.0338 (0.0026) | 0.0736 (0.0037) | 0.0314 (0.0025) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0396 (0.0028) | 0.0538 (0.0032) | 0.0194 (0.0020) |
| 100     | 0.0000 (0.0000) | 0.0424 (0.0028) | 0.0626 (0.0034) | 0.0264 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0678 (0.0036) | 0.0320 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0474 (0.0030) | 0.0578 (0.0033) | 0.0282 (0.0023) |
| 1000    | 0.0000 (0.0000) | 0.0338 (0.0026) | 0.0736 (0.0037) | 0.0314 (0.0025) |

Table 22: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.1$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0392 (0.0027) | 0.0496 (0.0031) | 0.0424 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0502 (0.0031) | 0.0580 (0.0033) | 0.0390 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0400 (0.0028) | 0.0620 (0.0034) | 0.0430 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0474 (0.0030) | 0.0554 (0.0032) | 0.0382 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0398 (0.0028) | 0.0618 (0.0034) | 0.0464 (0.0030) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0392 (0.0027) | 0.0496 (0.0031) | 0.0424 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0502 (0.0031) | 0.0580 (0.0033) | 0.0390 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0400 (0.0028) | 0.0620 (0.0034) | 0.0430 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0474 (0.0030) | 0.0554 (0.0032) | 0.0382 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0398 (0.0028) | 0.0618 (0.0034) | 0.0464 (0.0030) |

Table 23: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.1$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0532 (0.0032) | 0.0394 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0410 (0.0028) | 0.0638 (0.0035) | 0.0376 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0348 (0.0026) | 0.0666 (0.0035) | 0.0358 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0428 (0.0029) | 0.0612 (0.0034) | 0.0398 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0272 (0.0023) | 0.0634 (0.0034) | 0.0374 (0.0027) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0532 (0.0032) | 0.0394 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0410 (0.0028) | 0.0638 (0.0035) | 0.0376 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0348 (0.0026) | 0.0666 (0.0035) | 0.0358 (0.0026) |
| 500     | 0.0000 (0.0000) | 0.0428 (0.0029) | 0.0612 (0.0034) | 0.0398 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0272 (0.0023) | 0.0634 (0.0034) | 0.0374 (0.0027) |

Table 24: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.1$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0378 (0.0027) | 0.0602 (0.0034) | 0.0230 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0398 (0.0028) | 0.0660 (0.0035) | 0.0218 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0314 (0.0025) | 0.0726 (0.0037) | 0.0304 (0.0024) |
| 500     | 0.0000 (0.0000) | 0.0356 (0.0026) | 0.0640 (0.0035) | 0.0298 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0306 (0.0024) | 0.0744 (0.0037) | 0.0250 (0.0022) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0378 (0.0027) | 0.0602 (0.0034) | 0.0230 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0398 (0.0028) | 0.0660 (0.0035) | 0.0218 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0314 (0.0025) | 0.0726 (0.0037) | 0.0304 (0.0024) |
| 500     | 0.0000 (0.0000) | 0.0356 (0.0026) | 0.0640 (0.0035) | 0.0298 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0306 (0.0024) | 0.0744 (0.0037) | 0.0250 (0.0022) |

Table 25: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.1$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0286 (0.0024) | 0.0694 (0.0036) | 0.0204 (0.0020) |
| 100     | 0.0000 (0.0000) | 0.0282 (0.0023) | 0.0686 (0.0036) | 0.0210 (0.0020) |
| 200     | 0.0000 (0.0000) | 0.0278 (0.0023) | 0.0772 (0.0038) | 0.0284 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0332 (0.0025) | 0.0686 (0.0036) | 0.0224 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0300 (0.0024) | 0.0780 (0.0038) | 0.0266 (0.0023) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0286 (0.0024) | 0.0694 (0.0036) | 0.0204 (0.0020) |
| 100     | 0.0000 (0.0000) | 0.0282 (0.0023) | 0.0686 (0.0036) | 0.0210 (0.0020) |
| 200     | 0.0000 (0.0000) | 0.0278 (0.0023) | 0.0772 (0.0038) | 0.0284 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0332 (0.0025) | 0.0686 (0.0036) | 0.0224 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0300 (0.0024) | 0.0780 (0.0038) | 0.0266 (0.0023) |

Table 26: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.2$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0478 (0.0030) | 0.0574 (0.0033) | 0.0484 (0.0030) |
| 100     | 0.0000 (0.0000) | 0.0448 (0.0029) | 0.0552 (0.0032) | 0.0382 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0414 (0.0028) | 0.0630 (0.0034) | 0.0374 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0474 (0.0030) | 0.0570 (0.0033) | 0.0392 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0680 (0.0036) | 0.0402 (0.0028) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0478 (0.0030) | 0.0574 (0.0033) | 0.0484 (0.0030) |
| 100     | 0.0000 (0.0000) | 0.0448 (0.0029) | 0.0552 (0.0032) | 0.0382 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0414 (0.0028) | 0.0630 (0.0034) | 0.0374 (0.0027) |
| 500     | 0.0000 (0.0000) | 0.0474 (0.0030) | 0.0570 (0.0033) | 0.0392 (0.0027) |
| 1000    | 0.0000 (0.0000) | 0.0364 (0.0026) | 0.0680 (0.0036) | 0.0402 (0.0028) |

Table 27: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.2$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0414 (0.0028) | 0.0646 (0.0035) | 0.0372 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0404 (0.0028) | 0.0686 (0.0036) | 0.0352 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0678 (0.0036) | 0.0306 (0.0024) |
| 500     | 0.0000 (0.0000) | 0.0368 (0.0027) | 0.0660 (0.0035) | 0.0402 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0334 (0.0025) | 0.0700 (0.0036) | 0.0420 (0.0028) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0414 (0.0028) | 0.0646 (0.0035) | 0.0372 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0404 (0.0028) | 0.0686 (0.0036) | 0.0352 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0678 (0.0036) | 0.0306 (0.0024) |
| 500     | 0.0000 (0.0000) | 0.0368 (0.0027) | 0.0660 (0.0035) | 0.0402 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0334 (0.0025) | 0.0700 (0.0036) | 0.0420 (0.0028) |

Table 28: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.2$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0348 (0.0026) | 0.0726 (0.0037) | 0.0374 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0362 (0.0026) | 0.0760 (0.0037) | 0.0294 (0.0024) |
| 200     | 0.0000 (0.0000) | 0.0334 (0.0025) | 0.0764 (0.0038) | 0.0262 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0376 (0.0027) | 0.0750 (0.0037) | 0.0316 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0254 (0.0022) | 0.0782 (0.0038) | 0.0330 (0.0025) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0348 (0.0026) | 0.0726 (0.0037) | 0.0374 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0362 (0.0026) | 0.0760 (0.0037) | 0.0294 (0.0024) |
| 200     | 0.0000 (0.0000) | 0.0334 (0.0025) | 0.0764 (0.0038) | 0.0262 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0376 (0.0027) | 0.0750 (0.0037) | 0.0316 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0254 (0.0022) | 0.0782 (0.0038) | 0.0330 (0.0025) |

Table 29: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.5$ ,  $\eta = 0.2$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0368 (0.0027) | 0.0774 (0.0038) | 0.0318 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0334 (0.0025) | 0.0812 (0.0039) | 0.0310 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0280 (0.0023) | 0.0910 (0.0041) | 0.0250 (0.0022) |
| 500     | 0.0000 (0.0000) | 0.0294 (0.0024) | 0.0822 (0.0039) | 0.0228 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0208 (0.0020) | 0.0986 (0.0042) | 0.0274 (0.0023) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0368 (0.0027) | 0.0774 (0.0038) | 0.0318 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0334 (0.0025) | 0.0812 (0.0039) | 0.0310 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0280 (0.0023) | 0.0910 (0.0041) | 0.0250 (0.0022) |
| 500     | 0.0000 (0.0000) | 0.0294 (0.0024) | 0.0822 (0.0039) | 0.0228 (0.0021) |
| 1000    | 0.0000 (0.0000) | 0.0208 (0.0020) | 0.0986 (0.0042) | 0.0274 (0.0023) |



Table 30: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.05$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0510 (0.0031) | 0.0492 (0.0031) | 0.0402 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0578 (0.0033) | 0.0474 (0.0030) | 0.0428 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0510 (0.0031) | 0.0594 (0.0033) | 0.0438 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0524 (0.0032) | 0.0432 (0.0029) |
| 1000    | 0.0000 (0.0000) | 0.0412 (0.0028) | 0.0602 (0.0034) | 0.0404 (0.0028) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0510 (0.0031) | 0.0492 (0.0031) | 0.0402 (0.0028) |
| 100     | 0.0000 (0.0000) | 0.0578 (0.0033) | 0.0474 (0.0030) | 0.0428 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0510 (0.0031) | 0.0594 (0.0033) | 0.0438 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0524 (0.0032) | 0.0432 (0.0029) |
| 1000    | 0.0000 (0.0000) | 0.0412 (0.0028) | 0.0602 (0.0034) | 0.0404 (0.0028) |

Table 31: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.05$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0596 (0.0033) | 0.0504 (0.0031) | 0.0320 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0600 (0.0034) | 0.0592 (0.0033) | 0.0334 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0498 (0.0031) | 0.0572 (0.0033) | 0.0332 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0536 (0.0032) | 0.0572 (0.0033) | 0.0398 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0450 (0.0029) | 0.0554 (0.0032) | 0.0348 (0.0026) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0596 (0.0033) | 0.0504 (0.0031) | 0.0320 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0600 (0.0034) | 0.0592 (0.0033) | 0.0334 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0498 (0.0031) | 0.0572 (0.0033) | 0.0332 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0536 (0.0032) | 0.0572 (0.0033) | 0.0398 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0450 (0.0029) | 0.0554 (0.0032) | 0.0348 (0.0026) |

Table 32: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.05$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0608 (0.0034) | 0.0550 (0.0032) | 0.0274 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0594 (0.0033) | 0.0582 (0.0033) | 0.0312 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0458 (0.0030) | 0.0606 (0.0034) | 0.0268 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0466 (0.0030) | 0.0504 (0.0031) | 0.0350 (0.0026) |
| 1000    | 0.0000 (0.0000) | 0.0454 (0.0029) | 0.0656 (0.0035) | 0.0388 (0.0027) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0608 (0.0034) | 0.0550 (0.0032) | 0.0274 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0594 (0.0033) | 0.0582 (0.0033) | 0.0312 (0.0025) |
| 200     | 0.0000 (0.0000) | 0.0458 (0.0030) | 0.0606 (0.0034) | 0.0268 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0466 (0.0030) | 0.0504 (0.0031) | 0.0350 (0.0026) |
| 1000    | 0.0000 (0.0000) | 0.0454 (0.0029) | 0.0656 (0.0035) | 0.0388 (0.0027) |

Table 33: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.05$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0642 (0.0035) | 0.0564 (0.0033) | 0.0228 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0604 (0.0034) | 0.0624 (0.0034) | 0.0272 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0516 (0.0031) | 0.0580 (0.0033) | 0.0274 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0522 (0.0031) | 0.0516 (0.0031) | 0.0282 (0.0023) |
| 1000    | 0.0000 (0.0000) | 0.0442 (0.0029) | 0.0618 (0.0034) | 0.0270 (0.0023) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0642 (0.0035) | 0.0564 (0.0033) | 0.0228 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0604 (0.0034) | 0.0624 (0.0034) | 0.0272 (0.0023) |
| 200     | 0.0000 (0.0000) | 0.0516 (0.0031) | 0.0580 (0.0033) | 0.0274 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0522 (0.0031) | 0.0516 (0.0031) | 0.0282 (0.0023) |
| 1000    | 0.0000 (0.0000) | 0.0442 (0.0029) | 0.0618 (0.0034) | 0.0270 (0.0023) |

Table 34: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.1$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0574 (0.0033) | 0.0496 (0.0031) | 0.0434 (0.0029) |
| 100     | 0.0000 (0.0000) | 0.0556 (0.0032) | 0.0576 (0.0033) | 0.0460 (0.0030) |
| 200     | 0.0000 (0.0000) | 0.0480 (0.0030) | 0.0554 (0.0032) | 0.0444 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0572 (0.0033) | 0.0512 (0.0031) | 0.0442 (0.0029) |
| 1000    | 0.0000 (0.0000) | 0.0456 (0.0030) | 0.0594 (0.0033) | 0.0418 (0.0028) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0574 (0.0033) | 0.0496 (0.0031) | 0.0434 (0.0029) |
| 100     | 0.0000 (0.0000) | 0.0556 (0.0032) | 0.0576 (0.0033) | 0.0460 (0.0030) |
| 200     | 0.0000 (0.0000) | 0.0480 (0.0030) | 0.0554 (0.0032) | 0.0444 (0.0029) |
| 500     | 0.0000 (0.0000) | 0.0572 (0.0033) | 0.0512 (0.0031) | 0.0442 (0.0029) |
| 1000    | 0.0000 (0.0000) | 0.0456 (0.0030) | 0.0594 (0.0033) | 0.0418 (0.0028) |

Table 35: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.1$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0564 (0.0033) | 0.0534 (0.0032) | 0.0284 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0544 (0.0032) | 0.0544 (0.0032) | 0.0348 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0510 (0.0031) | 0.0664 (0.0035) | 0.0310 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0512 (0.0031) | 0.0394 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0384 (0.0027) | 0.0570 (0.0033) | 0.0344 (0.0026) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0564 (0.0033) | 0.0534 (0.0032) | 0.0284 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0544 (0.0032) | 0.0544 (0.0032) | 0.0348 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0510 (0.0031) | 0.0664 (0.0035) | 0.0310 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0512 (0.0031) | 0.0394 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0384 (0.0027) | 0.0570 (0.0033) | 0.0344 (0.0026) |

Table 36: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.1$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0616 (0.0034) | 0.0562 (0.0033) | 0.0332 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0528 (0.0032) | 0.0542 (0.0032) | 0.0346 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0458 (0.0030) | 0.0596 (0.0033) | 0.0258 (0.0022) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0554 (0.0032) | 0.0300 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0440 (0.0029) | 0.0586 (0.0033) | 0.0298 (0.0024) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0616 (0.0034) | 0.0562 (0.0033) | 0.0332 (0.0025) |
| 100     | 0.0000 (0.0000) | 0.0528 (0.0032) | 0.0542 (0.0032) | 0.0346 (0.0026) |
| 200     | 0.0000 (0.0000) | 0.0458 (0.0030) | 0.0596 (0.0033) | 0.0258 (0.0022) |
| 500     | 0.0000 (0.0000) | 0.0548 (0.0032) | 0.0554 (0.0032) | 0.0300 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0440 (0.0029) | 0.0586 (0.0033) | 0.0298 (0.0024) |

Table 37: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.1$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0682 (0.0036) | 0.0510 (0.0031) | 0.0234 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0646 (0.0035) | 0.0586 (0.0033) | 0.0222 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0544 (0.0032) | 0.0688 (0.0036) | 0.0222 (0.0021) |
| 500     | 0.0000 (0.0000) | 0.0524 (0.0032) | 0.0580 (0.0033) | 0.0300 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0432 (0.0029) | 0.0612 (0.0034) | 0.0302 (0.0024) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0682 (0.0036) | 0.0510 (0.0031) | 0.0234 (0.0021) |
| 100     | 0.0000 (0.0000) | 0.0646 (0.0035) | 0.0586 (0.0033) | 0.0222 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0544 (0.0032) | 0.0688 (0.0036) | 0.0222 (0.0021) |
| 500     | 0.0000 (0.0000) | 0.0524 (0.0032) | 0.0580 (0.0033) | 0.0300 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0432 (0.0029) | 0.0612 (0.0034) | 0.0302 (0.0024) |

Table 38: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.2$ , and  $c = 0.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0468 (0.0030) | 0.0494 (0.0031) | 0.0460 (0.0030) |
| 100     | 0.0000 (0.0000) | 0.0564 (0.0033) | 0.0506 (0.0031) | 0.0450 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0438 (0.0029) | 0.0632 (0.0034) | 0.0424 (0.0028) |
| 500     | 0.0000 (0.0000) | 0.0562 (0.0033) | 0.0514 (0.0031) | 0.0400 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0608 (0.0034) | 0.0442 (0.0029) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0468 (0.0030) | 0.0494 (0.0031) | 0.0460 (0.0030) |
| 100     | 0.0000 (0.0000) | 0.0564 (0.0033) | 0.0506 (0.0031) | 0.0450 (0.0029) |
| 200     | 0.0000 (0.0000) | 0.0438 (0.0029) | 0.0632 (0.0034) | 0.0424 (0.0028) |
| 500     | 0.0000 (0.0000) | 0.0562 (0.0033) | 0.0514 (0.0031) | 0.0400 (0.0028) |
| 1000    | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0608 (0.0034) | 0.0442 (0.0029) |

Table 39: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.2$ , and  $c = 1.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0568 (0.0033) | 0.0514 (0.0031) | 0.0380 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0496 (0.0031) | 0.0582 (0.0033) | 0.0372 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0480 (0.0030) | 0.0604 (0.0034) | 0.0328 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0518 (0.0031) | 0.0548 (0.0032) | 0.0314 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0384 (0.0027) | 0.0614 (0.0034) | 0.0320 (0.0025) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0568 (0.0033) | 0.0514 (0.0031) | 0.0380 (0.0027) |
| 100     | 0.0000 (0.0000) | 0.0496 (0.0031) | 0.0582 (0.0033) | 0.0372 (0.0027) |
| 200     | 0.0000 (0.0000) | 0.0480 (0.0030) | 0.0604 (0.0034) | 0.0328 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0518 (0.0031) | 0.0548 (0.0032) | 0.0314 (0.0025) |
| 1000    | 0.0000 (0.0000) | 0.0384 (0.0027) | 0.0614 (0.0034) | 0.0320 (0.0025) |

Table 40: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.2$ , and  $c = 1.5$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0552 (0.0032) | 0.0554 (0.0032) | 0.0264 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0558 (0.0032) | 0.0612 (0.0034) | 0.0304 (0.0024) |
| 200     | 0.0000 (0.0000) | 0.0560 (0.0033) | 0.0682 (0.0036) | 0.0324 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0530 (0.0032) | 0.0528 (0.0032) | 0.0306 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0572 (0.0033) | 0.0314 (0.0025) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0552 (0.0032) | 0.0554 (0.0032) | 0.0264 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0558 (0.0032) | 0.0612 (0.0034) | 0.0304 (0.0024) |
| 200     | 0.0000 (0.0000) | 0.0560 (0.0033) | 0.0682 (0.0036) | 0.0324 (0.0025) |
| 500     | 0.0000 (0.0000) | 0.0530 (0.0032) | 0.0528 (0.0032) | 0.0306 (0.0024) |
| 1000    | 0.0000 (0.0000) | 0.0406 (0.0028) | 0.0572 (0.0033) | 0.0314 (0.0025) |

Table 41: Simulation VII: Empirical power and Monte Carlo standard error for  $x_0 = 0.75$ ,  $\eta = 0.2$ , and  $c = 2.0$ .

| $r = 2$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0540 (0.0032) | 0.0556 (0.0032) | 0.0270 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0554 (0.0032) | 0.0574 (0.0033) | 0.0226 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0470 (0.0030) | 0.0614 (0.0034) | 0.0264 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0440 (0.0029) | 0.0570 (0.0033) | 0.0266 (0.0023) |
| 1000    | 0.0000 (0.0000) | 0.0440 (0.0029) | 0.0648 (0.0035) | 0.0226 (0.0021) |

  

| $r = 5$ | Uniform         | Central         | Tail            | Localized       |
|---------|-----------------|-----------------|-----------------|-----------------|
| 50      | 0.0000 (0.0000) | 0.0540 (0.0032) | 0.0556 (0.0032) | 0.0270 (0.0023) |
| 100     | 0.0000 (0.0000) | 0.0554 (0.0032) | 0.0574 (0.0033) | 0.0226 (0.0021) |
| 200     | 0.0000 (0.0000) | 0.0470 (0.0030) | 0.0614 (0.0034) | 0.0264 (0.0023) |
| 500     | 0.0000 (0.0000) | 0.0440 (0.0029) | 0.0570 (0.0033) | 0.0266 (0.0023) |
| 1000    | 0.0000 (0.0000) | 0.0440 (0.0029) | 0.0648 (0.0035) | 0.0226 (0.0021) |